

Sec. 1.8 De Morgan's law for quantified statements

Negation of "for all" and "there exists":

- $\neg(\forall x, P(x)) \equiv$
- $\neg(\exists x, P(x)) \equiv$

Example 1. What is the negation of "There exist positive integers a and b such that $\left(\frac{a}{b}\right)^2 = 2$."?

- Original statement in symbol:
- Take negation and apply De Morgan's Law:

Example 2. The domain is a set of students at a university. Define the following predicates:

$E(x)$: x is enrolled in the class. $T(x)$: x took the test.

What is the negation of "Everyone who took the test is enrolled in the class."?

Summary - fill in the blanks. Answers will be posted on Canvas.

Original statement	Negation of statement	Example of original	Example of negation
$p \wedge q$		"2 is prime and 2 is even"	
$p \vee q$		" $2^3 = 6$ or $\sqrt[3]{4} = 2$ "	
$p \rightarrow q$		"If 2 is even, then 6 is odd. "	
$\exists x, P(x)$		"There is a real number x such that $x^2 = -1$ "	
$\forall x, P(x)$		"For all integer n , -1 is a factor of n "	